Project Insight: Cricbuzz LiveStats

1. Project Overview

Cricbuzz LiveStats Dashboard is a web application built using **Streamlit** and **MySQL** to manage and visualize cricket data, including teams, matches, and player statistics.
The application allows users to:

- Perform **CRUD operations** (Create, Read, Update) on teams.
- Execute **predefined SQL queries** to analyze cricket matches and player performance.
- View **live or recent match data** fetched from the Cricbuzz API.

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2. Technologies Used

Technology	Purpose
Python	Core programming language
Streamlit	Web interface for interactive dashboard
MySQL	Database to store teams, matches, and player data
Pandas	Data manipulation and display
Requests	API calls to fetch live cricket data
Cricbuzz API	Source of cricket matches and player statistics

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3. Database Schema

Teams Table

Column Name	Type	Key	Description
team_id	INT	PK	Unique ID for each team
team_name	VARCHAR(255)		Full name of the team
team_sname	VARCHAR(10)		Short name (e.g., IND)

Matches Table (simplified)

Column Name	Type	Description
match_id	INT	Unique match identifier
series_name	VARCHAR	Series name

Column Name	Type	Description
team1_id	INT	Foreign key to Teams
team2_id	INT	Foreign key to Teams
status	VARCHAR	Match status/result

Players Table

Column Name	Type	Description
player_id	INT	Unique player identifier
full_name	VARCHAR	Player name
playing_role	VARCHAR	Batsman/Bowler/All-rounder

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4. Features

A. Team CRUD Operations

- **Add Team:** Insert new team data into teams table.
- **Update Team:** Modify team name or short name.
- **View Teams:** Display all teams in a table format.

B. SQL Query Analytics

- Run **predefined SQL queries** for insights such as:
    - Matches played in last 30 days.
    - Top winning teams.
    - Matches won per team.
    - Player performance stats.
    - Head-to-head predictions.

C. Live Matches & Player Stats

- Fetch **live/recent matches** using the Cricbuzz API.
- Insert or update match, team, and player data into MySQL.
- Display performance stats (batting, bowling) and partnerships.

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6. Project Workflow

1. **Data Ingestion**
    - Fetch live matches from API.

- Insert teams, matches, and players into database.

2. **CRUD Operations**
    - Users can add or update teams.
    - Teams with existing matches cannot be deleted (optional restriction).
3. **Data Analytics**
    - Run predefined queries via dropdown menu.
    - Display results interactively.
4. **Visualization**
    - Use Streamlit’s st.dataframe() for tabular display.
    - Interactive forms for user input.

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10. Conclusion

This project demonstrates the integration of:

- **Streamlit** for rapid web application development.
- **MySQL** for robust data storage.
- **External APIs** for live data ingestion.
- **SQL analytics** for meaningful cricket insights.
